

Motion-Only Capsize Warning on an Audited Synthetic Benchmark: Ranking Skill Without Threshold Transfer

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Abstract

Can the measured roll motion of a ship warn of capsizing soon enough to act, at a false-alarm cost a crew could accept? We study this question on a synthetic benchmark validated against analytic limits and internal numerical checks; external ship-level validity is out of scope: a one-degree-of-freedom nonlinear roll model, three restoring-curve families representing distinct ways a ship loses stability, and a baseline detector corpus of 58,500 seeded trajectories under JONSWAP forcing, with later preregistered experiments materializing separate additional campaigns. Five warning methods compete under one protocol in which every operating threshold is chosen on calibration data and frozen before each test evaluation, with every historical deviation from that protocol recorded in the repository rather than repaired silently: a small convolutional network, classical variance and autocorrelation trend statistics, a likelihood-ratio detector, a closed-form critical-roll-rate margin, and a phase-space neighbor-count score. Four findings emerge. (a) Ranking skill is real at every forcing bandwidth tested; the fixed-window primary network separates dangerous from ordinary intervals with window AUC 0.883 to 0.913. (b) Inside an established severe regime, no tested motion-only statistic predicts which encounter will be fatal; all score near chance. (c) Most nominally false alarms, 75 to 88 percent in a mixed-generation historical diagnostic, coincide with high-envelope wave-group episodes, a coincidence proxy without a matched null, that the vessel survived. (d) No method's threshold transfers across failure modes at fixed sensitivity. We then operate the split-time upcrossing decomposition as an onboard rate estimator, in online and offline-calibrated forms; both fail preregistered calibration tests for a reason we localize. In an in-sample check, the decomposition reproduced realized counts: given the empirically measured probability that a threshold crossing is terminal, it reproduces realized counts within 3 in 71. But the closed-form criticality model overestimates the lethality of a crossing nearly fivefold, and in 92 percent of heralded capsize (221 of 240) no ten-second emission opportunity separated the terminal crossing from the recorded capsizing. An audit during the study found our own first results inflated by test-set threshold selection; every number reported here is the corrected value. The picture that survives is simple, and a final embedded-group experiment quantifies it within its own design. Embedding a frozen library of six wave-group shapes at six prescribed heights into independent irregular preludes, so that the vessel arrives at each group naturally, capsizes within the group is predicted at AUC 0.513 from the vessel's entry state, 0.934 from the group's engineered descriptors, and 0.938 from both. Within that balanced single-configuration experiment, motion reveals the slowly building vulnerability and the arriving group's description dominates the rest.

Highlights:

- Five motion-only capsizing scores were compared with frozen thresholds.
- Window AUC stayed 0.883 to 0.913 across tested forcing bandwidths.
- Tested scores were near chance inside an established severe regime.

- No 10-s emission opportunity preceded 92 percent of heralded capsizes.
- In one six-shape library, wave-group descriptors gave AUC 0.934.

Keywords: capsize, intact stability, real-time warning, split-time method, preregistered evaluation, synthetic benchmark.

1. Introduction

Begin from first principles. A ship in beam seas is a lightly damped nonlinear oscillator (Nayfeh and Mook 1979): waves push, the righting arm restores, and capsize is the escape of that oscillator over the rim of its potential well. Two different questions hide inside “can we warn?” One asks whether the well is becoming shallow — vulnerability, a slow process. The other asks when a particular group of waves will push the ship over — the encounter, a fast one. Every result in this paper answers one of those two questions, and the paper’s central finding is that motion measurement answers only the first.

Interest in warning of capsize from onboard motion measurement is at least two decades old. The Lyapunov-exponent programme of the mid-2000s (McCue and Troesch 2004, 2006) sought a precursor of capsize in the divergence of measured trajectories; the author’s thesis in that programme (Story 2009) found empirically that the useful signal lay not in estimated exponents but in the loss of phase-space neighbors, and reported a neighbor-count warning flag with encouraging lead times and an honestly documented false-alarm problem. The thread was not taken up. The signal-based detectors of Galeazzi et al. (2013, 2015) for parametric roll, with designed false-alarm probabilities and full-scale blind validation, remain the most complete fielded success of motion-based stability-failure detection, though onboard monitoring efforts are again active, for damaged ships (Mak et al. 2025) and naval vessels (Santiago Caamaño et al. 2025), and the field’s general machinery moved instead to offline probabilistic assessment (Belenky and Sevastianov 2007), of which the split-time method (Belenky, Weems and Lin 2016; Belenky et al. 2024) is the principal example, and to operational guidance derived from pre-computed simulation under the second-generation intact stability criteria (Bačkalov et al. 2016; IMO 2020, 2022; Petacco and Gualeni 2020; Shigunov 2023).

The relationship of the present work to the split-time method deserves statement at the outset, since that method supplies both the strongest available machinery and the sharpest open question. Belenky et al. (2024) compute the probability of capsizing offline. Their central object is an upcrossing: the roll angle passing outward through a fixed intermediate threshold, here the angle of maximum righting arm. The metric of danger, a critical roll rate at those upcrossings, is evaluated along simulated histories, its tail is extrapolated with a physics-informed exponential form (Glotzer et al. 2024), and the estimate is validated by confidence-interval capture against direct counting (Weems et al. 2023; on direct-counting interval estimands see Wandji et al. 2024). Three transfers from that programme are attempted here. The metric itself is operated, to our knowledge for the first time, as a continuously evaluated onboard alarm statistic computed from measured state alone, and is benchmarked against learned and statistical competitors on equal terms. The statistical discipline of the split-time programme, declustering, interval estimates on every rate, and validation against direct counting, is transplanted to warning evaluation, where it remains rare; an uncertainty-aware short-horizon warning benchmark has appeared in parallel (Joo et al. 2026). And the boundary of the transfer is itself measured:

the engineering-fidelity form of the metric requires re-simulation with the incident wave realization known, and the present results quantify operationally what that requirement conceals, namely that the information carried by the future waves is absent from measured motion precisely where event timing is decided.

The present study returns to the motion-only question with tools that did not exist in 2009 and with an evaluation protocol designed against the failure modes that make warning research easy to overstate. A warning method may rank dangerous intervals well yet possess no usable threshold; a threshold may fail to transfer across failure modes or sea states; apparent skill may arise from protocol time, truncated outcome windows, or the use of future data during normalization; and, because capsizes are rare, an aggregate score conceals all of these. Each failure mode is tested separately here. The study also reports, deliberately, an audit of its own methodology: the operating points first obtained were later found to have been selected on test outcomes, an error we believe is widespread in the published literature on learned motion prediction, and the corrected results quantify its size.

Three questions are addressed. First, does motion history rank capsize risk under forcing broad enough that classical critical-slowness indicators fail? Second, do the resulting operating thresholds transfer across stability-failure modes at fixed sensitivity? Third, what do the false alarms of such systems consist of?

2. The synthetic benchmark

The benchmark is the simplest ship that can capsize (Figure 1): a one-degree-of-freedom non-linear roll model, validated against analytic limits, generating seeded trajectories under JONSWAP forcing (Hasselmann et al. 1973). Three restoring families represent failure archetypes: softening restoring (dead-ship escape), time-varying stiffness (parametric roll), and biased restoring with asymmetric escape angles (damage, steady heel), in the scope of established one-degree-of-freedom roll comparisons (Bulian and Francescutto 2004, 2011). Following-sea failure modes, broaching foremost (Umeda et al. 2016), are outside the benchmark's scope, a limitation the transfer results of Section 5 should be read against. Forcing is synthesized spectrally with deterministic amplitudes and seeded random phases (Shinozuka and Deodatis 1991); integration is fixed-step Runge-Kutta at no fewer than 40 steps per natural period; capsize is an absorbing state and post-event samples are excluded from every window. The benchmark takes its name from Rahola (1939), whose thesis distilled casualty records into judged stability criteria; the aim here is the same empirical discipline applied to warning.

The model is nondimensional, and its forcing deserves a dimensional account. The excitation represents an effective wave-slope moment process: a JONSWAP elevation spectrum is converted to a wave-slope spectrum, scaled by an effective-wave-slope coefficient and the square of the natural frequency, and applied as a roll moment — the linear-limit validation gate below checks exactly this chain against the spectral calculation. The natural period (four seconds at reference), damping ratio, and escape-angle geometry are set to representative small-vessel values; the three restoring shapes and the effective-wave-slope coefficient are chosen for archetypal behavior, not calibrated to a hull. No coefficient in the benchmark is claimed as a measured property of any ship, which is one reason external ship-level validity is declared out of scope.

Validation gates precede all warning experiments: recovered significant wave height within 2 percent; linear-limit response variance within 6 percent of the spectral calculation; the prin-

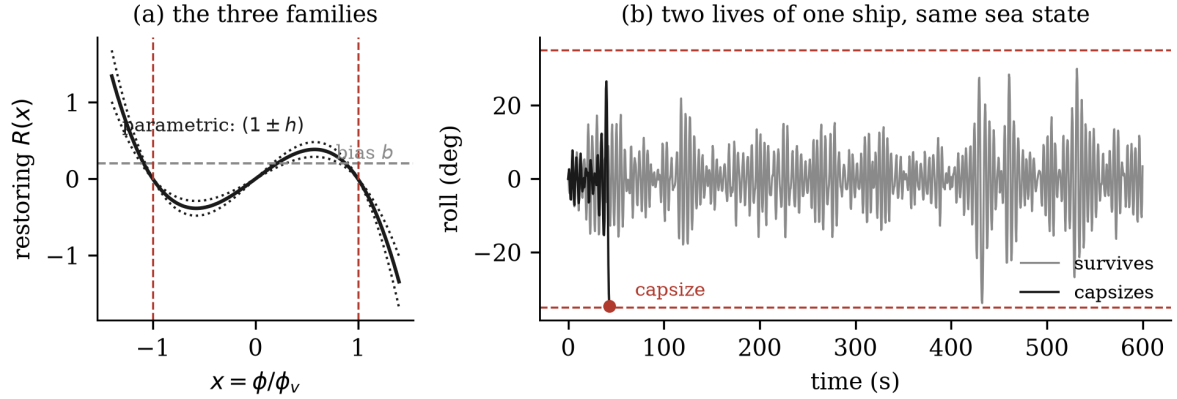


Figure 1: The benchmark. (a) The three restoring families: the softening curve with its escape angles (dashed red), the constant bias moment that shifts the biased family’s equilibria, and the parametric family’s stiffness modulation envelope (dotted). (b) Two 600-second lives of the same ship in the same sea state, different seeded seas: the capsizing life ends during a wave group in its first minute, while its twin survives larger excursions for the full record. The encounter, not the general severity, decides the moment.

cipal Mathieu boundary within 10 percent of the averaging result; and the simulated harmonic capsize boundary at or above the heteroclinic Melnikov threshold (Falzarano, Shaw and Troesch 1992) with the correct frequency shape, the bound being treated as a necessary condition only. The acceptance run at the submission commit passes the full suite of 244 tests with none skipped, and the attained residuals of each gate are serialized in a versioned artifact (`results/validation_acceptance.json`) alongside their thresholds.

The baseline detector corpus comprises 58,500 trajectories of 600 seconds at a four-second natural period: stationary campaigns per family, stiffness-ramp campaigns representing slow stability erosion, one sea-state step campaign, rare-event evaluation campaigns at 0.95 to 2.0 percent capsize rates, and five forcing-bandwidth campaigns in which JONSWAP peak enhancement is varied from 1.0 to 30 with terminal ramp severity retuned per bandwidth so that outcome rates remain in a common band; the last decision prevents severity from acting as a hidden bandwidth axis. Disjoint seed blocks separate training, calibration, test, and guarded holdout data. The later preregistered experiments of Section 5 — the split-time operation, the tangent-exponent trial, the wave-field audit, and the embedded-group experiment — materialized separate additional campaigns, each recorded against the same seed ledger.

Supervised windows use 60 natural periods of causally normalized roll and roll-rate history, a 50-period outcome horizon, and an exclusion band for near misses; a future-only leakage probe verifies the entire feature path against a deliberately leaky control. The label discipline responds directly to critiques of early-warning-signal studies in other fields (Boettiger and Hastings 2012; Dakos et al. 2012).

3. Five warning methods

All five methods consume identical windows and emit a scalar score which is thresholded into alarm episodes (three consecutive crossings to open; refractory closure). No method receives wave elevation, spectrum, family identity, or sea-state input.

(a) A temporal convolutional network of 2,969 parameters trained on the horizon labels, in the spirit of learned early-warning classifiers (Bury et al. 2021); neural roll prediction has in-community precedent (Míguez González et al. 2023). (b) Rolling variance and lag-one autocorrelation with Kendall trend statistics, the classical critical-slowness indicators (Dakos et al. 2012), whose rate-of-change and noise-model limits are themselves documented (Radhakrishnan et al. 2025; Layritz et al. 2025). (c) A roll-power adaptation of the double-Weibull generalized likelihood-ratio detector of Galeazzi et al. (2013, 2015). (d) The split-time critical roll rate (Belenky et al. 2024), computed in closed form from the known restoring parameters and the measured state, with the separatrix line extrapolated between threshold crossings; to our knowledge this quantity has not previously been operated as a continuously evaluated online alarm statistic, and it functions here as a zero-training physics baseline. (e) The neighbor-count score of Story (2009), reimplemented faithfully in the (roll, roll-rate) plane under strictly causal normalization, with the thesis’s fixed flag retained as one point on a full operating curve; phase-space identification of dangerous ship states has since been developed independently in the Spyrou school (Kontolefas and Spyrou 2020).

4. Evaluation protocol and the audit

Metrics are episode sensitivity, false episodes per exposure hour, and lead time, with trajectory-block bootstrap intervals and fixed campaign weights; exposure ends at capsizes; an episode overlapping the pre-capsize horizon is event-associated. Final thresholds for the baseline detector experiments were selected on calibration data and frozen before the final corrected rescaling; those development-test blocks had already been accessed during method correction, so the baseline results are corrected retrospective evaluations, not one-shot tests. The later split-time, tangent-exponent, wave-field, and embedded-group experiments used separately recorded preregistered fresh-data procedures, detailed with each result.

The protocol was not followed from the outset, and the failure is reported as a result. The study’s first operating points were obtained by scanning test-set metrics for the lowest false-episode rate at 90 percent sensitivity or better, which is threshold selection on test outcomes. Reported false-episode rates under test-informed selection were several times lower than the calibration-frozen rates reported after correction; because the affected evaluations differ in population as well as threshold provenance, we report the effect qualitatively rather than as a single controlled factor. One guarded holdout evaluation was additionally invalidated by threshold selection on the holdout labels, and its artifacts are retained as historical records only. The corrected protocol is encoded in guard tests, and the acceptance run recorded with the validation artifact passes the full suite with none skipped. We suggest that reported operating points in the learned motion-prediction literature should be read with this mechanism in mind wherever threshold provenance is not stated.

Because the study accumulated experiments with different data-access histories, Table 1 states, for each, the primary question, the data it was evaluated on, how its operating choices were

selected, what its intervals mean, and its preregistered verdict. Throughout Section 5 each experiment is named by what it measures; the parenthesized labels (D1 through D5, U1, H1, F1, W1, D4b) are the repository identifiers under which every artifact, preregistration, and dated addendum is filed.

Table 1: The experiment ledger. Every experiment’s primary question, the data it was evaluated on, how its operating choices were selected, what its intervals mean, and its verdict. Confirmatory preregistered gates are distinguished from descriptive diagnostics; the D4 row is descriptive on both counts.

Experiment	Primary question	Data access	Selection rule	Intervals	Verdict
D1 pooled operating points	sensitivity vs. false-episode cost	corrected development-test rescoring	calibration-frozen thresholds	trajectory-block bootstrap	descriptive
D2 threshold transfer	does a frozen threshold transfer across families	corrected development-test rescoring	calibration-frozen thresholds	trajectory-block bootstrap	failed at fixed sensitivity
D3 bandwidth sweep	does ranking survive broadband forcing	corrected development-test rescoring	preregistered 10% operating gate	trajectory-block bootstrap	AUC held; operating gate missed
D4 false-alarm character	what false episodes coincide with	mixed-generation historical diagnostic	envelope criterion, predeclared	none (descriptive)	descriptive; no matched null
D5 within-regime timing	does motion time capsize inside the severe regime	corrected development-test rescoring	preregistered 0.58 trigger	trajectory-block bootstrap	chance-level
U1/U1-r2 online split-time	is the online decomposition calibrated	preregistered fresh test	preregistered capture criterion	parameter-uncertainty capture	failed; error localized
H1 offline-calibrated hybrid	is the offline-calibrated form calibrated	preregistered fresh test, one shot	preregistered capture and kill criteria	parameter-uncertainty capture diagnostic; reliability MAE	failed capture
F1 tangent exponents	do finite-time tangent statistics time capsize	preregistered, oracle and causal	preregistered ablation	trajectory-block bootstrap; survivor-conditioned predictive	chance-level for tested family
W1 wave-field audit	is the synthetic seaway statistically sound	preregistered fresh wave-only seeds	preregistered gates		passed except one continuous-Rice gate
D4b embedded groups	how is predictability split between entry state and group	preregistered fresh preludes, one shot	preregistered predictions, frozen library	prelude-clustered percentile bootstrap	C5 composition failed; C6 evaluated

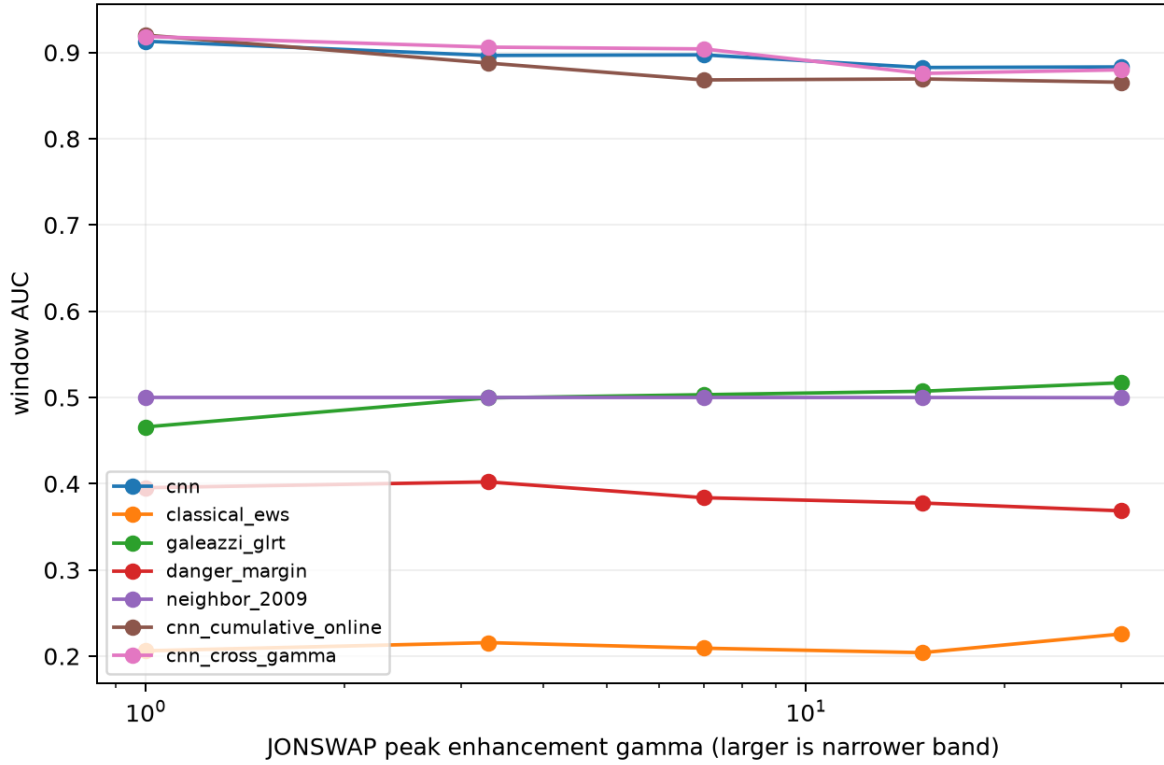


Figure 2: Ranking survives bandwidth. Window AUC on the five forcing-bandwidth campaigns, corrected record: the fixed-window primary network holds 0.883 to 0.913 from narrow-band forcing down to fully broadband, while the classical indicators and physics scores range from near chance to well below it on these ramped campaigns — the fixed-window scoring that is honest about information also strips the trends those baselines rely on.

5. Results

Ranking under changing forcing bandwidth (D3). Classical critical-slowness indicators depend on narrow-band structure, so the first question is whether any motion score ranks risk once that structure is removed. Five campaigns vary the JONSWAP peak-enhancement factor from 30, a nearly monochromatic sea, down to 1.0, fully broadband; every method scores the same windows, and window AUC measures how well each score ranks the intervals that end badly above those that do not. The fixed-window primary network’s window AUC was 0.883 to 0.913 at every peak enhancement from 30 down to 1.0, while the classical trend indicators collapsed on ramped campaigns. Ranking skill from motion history survives broadband forcing. The predeclared operating-cost criterion at the broadband end (a 10 percent false-episode improvement) was missed at 7.6 percent (14.061442 versus 15.212539 false episodes per hour), and the operating-cost claim is reported as inconclusive: ranking skill and a deployable threshold are different assets (Figure 2).

Operating points at frozen thresholds (D1). Ranking skill is not an alarm. Here each method’s score stream is thresholded into alarm episodes — three consecutive crossings open an episode,

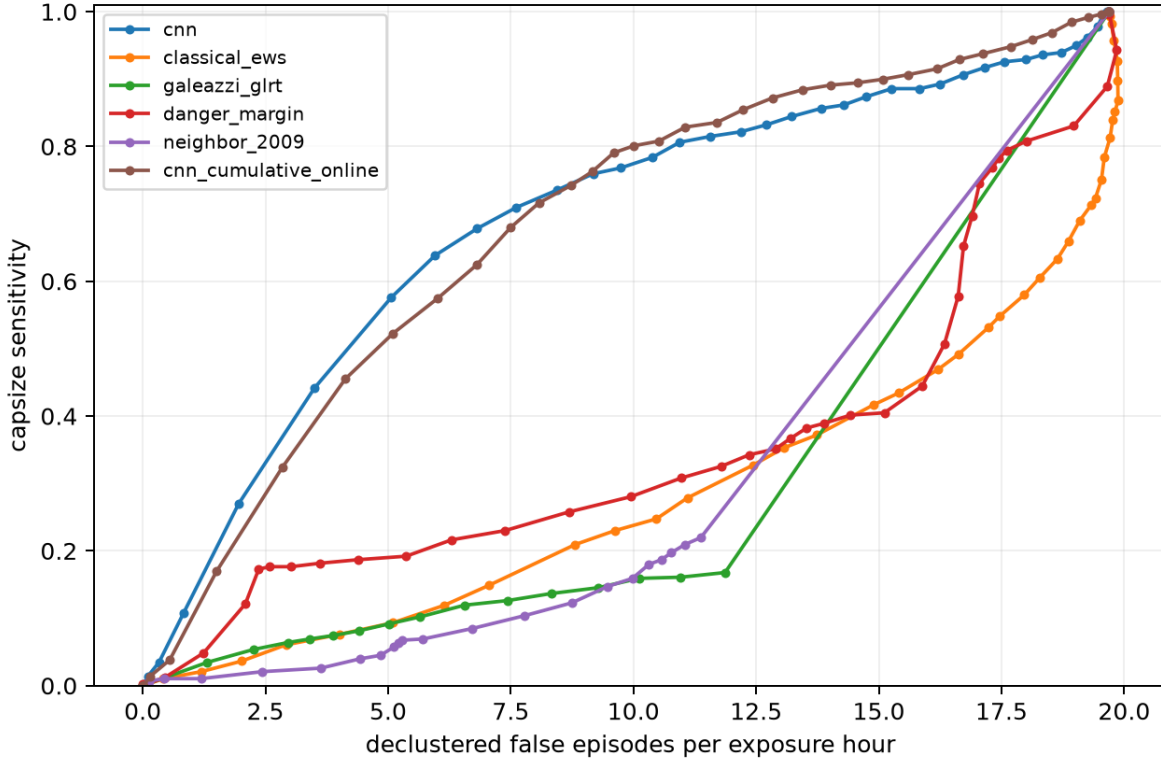


Figure 3: The pooled operating curves, corrected record: capsizes sensitivity against declustered false episodes per exposure hour at calibration-frozen thresholds. The network variants dominate the low-cost region; every classical and physics score buys sensitivity only near the always-on corner. Even the best curve pays several false episodes per hour at 90 percent sensitivity.

a refractory period closes it — with the threshold chosen on calibration data and frozen, and the resulting alarm system is charged its full operating cost: episode sensitivity against false episodes per exposure hour on the pooled rare-event campaigns. At calibration-frozen thresholds the fixed-window primary network attained 91.00 percent sensitivity at 13.409 false episodes per hour; the classical indicators reached 100 percent sensitivity only at a near-always-on threshold costing 21.4 per hour; the physics margin and the 2009 neighbor score occupied the same high-sensitivity, high-cost region. No method reached 90 percent sensitivity at a false-episode rate below several per hour (Figure 3).

Threshold transfer across failure modes (D2). A deployed system cannot know which failure mode the ship will face, so a threshold chosen on one pair of restoring families is applied unchanged to the held-out third, in rotation, and judged at the same fixed sensitivity target. The fixed-window CNN reached 99.15 percent, 90.83 percent, and 76.21 percent test sensitivity across the softening, parametric, and biased holdouts. Two rotations met 90 percent, but only near always-on alarm cost; the biased holdout failed. No deployable transfer was established.

Timing inside the severe regime (D5). The two clocks of the Introduction are separated by construction: evaluation is restricted to trajectories already washed into the established severe

regime, where the slow vulnerability signal has saturated, so any remaining skill must come from reading the fast clock — which particular encounter will be terminal. After the full wash-in, the fixed-window CNN’s orientation-independent AUC was 0.556, below the 0.58 trigger; cumulative-online CNN reached 0.538 and classical scores remained near chance. Motion-only scores carried no usable information about which encounter would be terminal inside the established severe regime.

The character of false alarms (D4). What is a false alarm made of? Because the seaway is synthetic and fully known, every false episode can be checked for coincidence with a high-envelope wave group, defined by a Hilbert-envelope exceedance criterion held for a minimum run of cycles. Between 75 and 88 percent of nominally false episodes coincided with such groups, which the vessel survived. This is a coincidence proxy, not the critical-wave-groups method proper (Themelis and Spyrou 2007; Anastopoulos et al. 2016). The pooled range also mixes artifact generations: the danger-margin row is v0.2-regenerated while the remaining rows are v0.1-corrected, as the repository’s supersession tables record. The figure is descriptive, since no matched null was tested and groups are common in a narrow-band sea; its qualitative content is that outcome-based false-positive accounting charges the detectors for hazard exposure the vessel happened to survive, the base-rate structure anticipated in Boettiger and Hastings (2012).

Reference comparators. To locate the learned methods, three references were scored on a design-balanced window set: a restart comparator that re-simulates from the window’s exact end state under independent future forcing, an upper reference for state information alone; a gradient-boosted tree on engineered features, among them an online stiffness estimate; and the fixed-window network itself. The exact-state independent- future restart reached AUC 0.850, fixed-window XGBoost 0.723, and fixed-window CNN 0.652; a protocol-time comparator also exceeded the network. Two qualifications apply: the restart comparators replace the realized forcing and are therefore reference points rather than bounds, since the temporally correlated seaway leaves encounter information in the motion history; and part of the network’s advantage on natural test populations evidently derived from population structure rather than state inference. The headroom that exists lies in state inference, and simple tools claimed it first.

Operating the split-time decomposition onboard (U1, H1). The split-time method writes the capsized rate as a product: the rate at which roll upcrosses an intermediate threshold, times the probability that an upcrossing goes critical. Three preregistered experiments operated that product as a continuously updated onboard estimate computed from measured motion alone: one with the conditional estimated online from the observed exceedance tail (U1), one attributing the resulting error term by term against measured quantities (U1-r2), and one with the conditional calibrated offline from design-stage data and multiplied onboard by the measured crossing rate (H1). Each was judged on fresh test data by the split-time program’s own confidence-interval-capture standard: does the estimate’s interval capture the realized capsized count of each campaign? The online-estimated form failed its capture criterion, and a predeclared comparison found the upcrossing rate alone more reliable than the full decomposition, so the corresponding kill criterion fired. A measured attribution then localized the error: with the empirical terminal-crossing probability substituted for the modeled one, the decomposition predicted 68 events against 71 realized, while the closed-form criticality model (the unforced piecewise-linear critical roll rate with an exponential tail) overpredicted by 4.8-fold; a motion-derived estimate of the forced correction made the error larger, not smaller. A final hybrid,

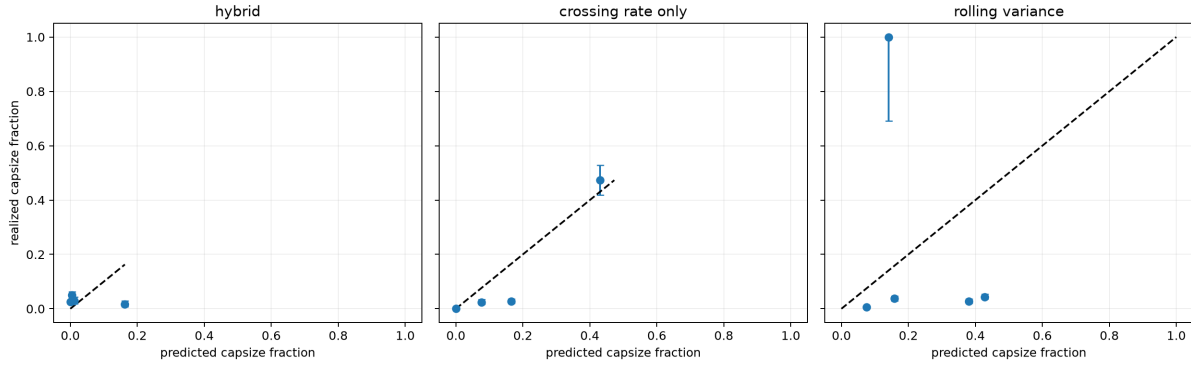


Figure 4: Reliability of the offline-calibrated hybrid and its comparators on fresh test data (H1): predicted capsizes probability per bin against the realized fraction. All three methods failed campaign count capture. The hybrid and rate-only maps show similar pooled reliability error (0.0398 and 0.0428) and both outperform the variance map (0.2423); no method was validated as a calibrated operational estimator.

with the conditional calibrated offline from design-stage data and multiplied onboard by the measured crossing rate, also failed its capture and transfer criteria. A timing audit supplied the mechanism: in 221 of 240 heralded capsizes (92 percent; 19 exceptions), no ten-second emission opportunity lay between the retained terminal crossing and the recorded capsizes — our reading is that the severity information the conditional carries arrives too late, on this emission grid, to act on. Two methodological qualifications attach: fresh test slices of 30 to 60 expected events realized counts 21 to 37 percent from expectation, and the capture intervals carried parameter uncertainty without realization noise — a frozen parameter-uncertainty diagnostic rather than calibrated predictive coverage — so the capture criterion on slices of this size is severe even for a correctly calibrated estimator; the reliability errors (0.039848 for the hybrid, 0.042806 for the rate-only map, and 0.242303 for a variance-based map) are the fairer calibration comparison. All verdicts were preregistered and stand as recorded. Figure 4 shows the reliability comparison: the hybrid and rate-only maps had similar pooled reliability error and both outperformed the variance map, but no method passed the campaign count-capture criterion, and neither form was validated as a calibrated operational estimator.

The exponent’s clean trial (F1). The 2009 thread deserved one experiment its era could not run: the finite-time Lyapunov exponent computed exactly, from the model’s analytic Jacobian, with tangent perturbations receiving the identical realized forcing (the standard construction of Babaee et al. 2017). One preregistered ablation evaluated the margin level, its closure rate, time to closure, energy reserve and depletion, level-plus-rate combinations, the instantaneous strain normal to the escape boundary, and generic and escape-directed finite-time exponents, in both an operational causal setting and an oracle setting with the true state and tangent map. On the fully post-step severe-regime geometry, every statistic scored between 0.501 and 0.507 — including the acausal true-tangent exponents, which use information from the future itself. The conditional plan to audit the 2009 estimator’s implementation was therefore retired by its own trigger: within the tested family of statistics, the exactly computed tangent quantities carried no timing information, so any historical implementation defect was immaterial to the thesis’s conclusion. The estimand deserves exactness. The evaluated population is 27,290

severe-regime windows (19,202 nonterminal, 8,088 terminal); finite-time scores are evaluated only on windows whose trajectory survives the tangent horizon, which censors 191, 419, and 1,018 terminal windows (and no nonterminal ones) at horizons of one, two, and five natural periods — a censoring that preferentially removes the fastest events. The generic statistic is the finite-time maximum singular value of the tangent propagator Φ ; the escape-directed statistic is the covector projection $\|n_0^T \Phi\|$ onto the initial-state escape normal, normalized over the horizon — an escape-normal finite-time tangent amplification, not the conventional singular-value exponent (Shadden et al. 2005). On this survivor-conditioned, fully post-step geometry, none of the tested finite-time tangent statistics discriminated terminal from nonterminal windows; no information-theoretic ceiling is computed or claimed.

The other side of the decomposition (D4b). A final experiment asked the question every negative above implies: if the arriving wave group were known, how much would it decide? A frozen library of six envelope-mined group shapes was embedded into hundreds of independent irregular preludes, in the natural-initial-condition manner of Silva and Maki (2021, 2024; extended to free-running vessels in Silva and Maki 2026), so that the vessel and its response history arrive at each group naturally; group heights were swept across the critical range and monotone response maps fitted per entry-state stratum. Evaluated once on held-out preludes (7,164 trials, 580 capsizes), capsize within the group was predicted at AUC 0.513 [0.503, 0.524] from the vessel’s entry state alone — near-chance, statistically above one half but operationally negligible — at 0.934 [0.930, 0.938] from the group’s engineered descriptors alone, and at 0.938 [0.934, 0.941] from both, with intervals from prelude-level bootstrap resampling. The three predictors are fixed-penalty L2-regularized logistic models on entry-state features, group descriptors, and their union, fitted on calibration preludes and evaluated once on held-out preludes; over the six-shape library the group and combined designs are rank-deficient, so individual coefficients are not identifiable, though in-support predictions are stable under the fixed penalty. The monotone response maps serve the rate composition below, not these AUCs. The preregistered expectation that the combination would substantially exceed either input was honestly not confirmed: the point estimate rises by 0.004 once the group is known, a difference for which no paired interval is reported. The estimand deserves precision: this is a balanced design over six frozen shapes and six prescribed heights, in-support (shapes and heights are not held out; prelude seeds are), for one softening-family configuration in beam seas, and the group descriptors include the imposed height — the experimentally manipulated driver. Within that design, group descriptors dominate the entry state; no universal division of predictability follows. A companion rate composition over the library’s naturally occurring groups failed its capture criterion in the informative direction: the composition predicted zero capsizes while direct counting on matched campaigns recorded 39 in 1,500 trials — every library class fell below its critical height. This six-class implementation failed coverage of the direct capsize mechanism; the frozen artifacts do not identify which omitted representation — sequences outside the mined classes, the entry-state stratification, or the fitted boundary — is responsible, and no conclusion about other libraries or parameterizations follows. Broadband group representations with initial-condition variability (Hafezi et al. 2026; cf. Anastopoulos and Spyrou 2019, 2023) indicate what a broader library must include. One numerical limitation attaches: a step-halving audit on a frozen 128-seed subset changed one capsize classification, so exact counts and rates are grid-conditioned rather than grid-independent constants, though no headline comparison could be reversed by a sensitivity of that size. A preregistered wave-field audit (W1) preceded these runs: the periodic record’s recurrence begins beyond every

labeled lag, a check the self-repetition discussion of Umeda et al. (2025) motivates; elevation upcrossing rates match the Rice (1945) formula in a passing-rate sense, though one preregistered continuous-Rice mean gate failed as a consequence of the fixed-energy conditioning; and that conditioning of the deterministic-amplitude ensemble is stated as part of the estimand.

A companion study of conformalized forecast intervals on the same benchmark (following Romano, Patterson and Candès 2019; Gibbs and Candès 2021; sequential and nonexchangeable variants in Zaffran et al. 2022, Xu and Xie 2023, and Barber et al. 2023) found stationary marginal coverage accurate to 0.75 percentage points in the mean, while every online adaptation scheme tested failed a predeclared joint rolling-coverage and alarm-cost criterion through the sea-state step. Those results are reported in the project record and not further discussed here.

6. Discussion

The results organize into a two-clock description. The slowly varying stability state, the eroding restoring that determines whether the vessel is in danger, is observable from motion history: ranking skill survives every forcing bandwidth, and the most informative engineered feature is in effect an online stiffness estimate, a quantity this community has long extracted from the roll period (Terada et al. 2016; Míguez González et al. 2017; full-scale voyage evidence in Higo et al. 2025). For the tested family of motion-only statistics, the terminal encounter, the particular wave group that converts vulnerability into capsizes, was not recovered once the severe regime was established; the within-regime result is chance-level for learned, classical, and physical scores alike, consistent with the stochastic-Melnikov view that a deterministic separatrix loses its predictive meaning under random forcing (Frey and Simiu 1993). The practical reading is that motion-only systems should be understood, and evaluated, as vulnerability monitors rather than event predictors, a role compatible with the operational measures framework of the second-generation criteria (IMO 2020) rather than with an alarm bell.

Read against the split-time programme, the results locate both the value and the limit of operating its metric onboard. As a zero-training detector the closed-form critical roll rate was sensitivity-dominant and cost-competitive with every learned alternative, which is at once a tribute to the physics and a caution against claims for learned detectors that have not faced it as a baseline. Its one systematic failure, degradation on stability-erosion campaigns where the assumed restoring becomes stale, identifies the missing component for operational use: on-line estimation of the current restoring state, whose feasibility the comparator results demonstrate, the roll period alone recovering most of the available headroom; full-scale data-driven parameter updating is developing in exactly this direction (Takami et al. 2026). The within-regime chance result then explains why the engineering-fidelity metric must re-simulate with the waves known: the future-dependent quantity used by the motion perturbation method was not recovered by the tested motion-only statistics. The U1 and H1 experiments made that boundary quantitative from inside the method itself. Among the tested components, the crossing rate — an observable of the slowly varying stability state — retained the more encouraging qualitative reliability pattern, but the rate-only map failed every campaign count-capture check and was not validated as a calibrated operational estimator; the severity factor, which would time the event, was in 92 percent of heralded capsizes not emitted with a ten-second opportunity before the recorded event. The two-clock description is therefore not an external criticism

of the split-time programme but a property its own decomposition exhibits when operated on-line: the observable factor reads the slow clock, while the tested online severity factor requires future re-simulation.

The false-alarm finding bears on how such monitors should be judged, though it supports a hypothesis rather than a conclusion. The high overlap between false episodes and high-envelope wave groups motivates hazard-conditioned evaluation, but without a prevalence-, duration-, and regime-matched null it cannot distinguish hazard sensitivity from the background prevalence of envelope groups in narrow-band forcing; whether conventional false-alarm accounting is biased, and in which direction, awaits that matched-null experiment. If the overlap survives it, metrics conditioned on hazard exposure, and alarm policies whose budgets are conditioned on regime — designed within the bridge alert-management framework (IMO 2010), with decision support under uncertainty developing in parallel (Louvros et al. 2025) — appear to us a more productive direction than further detector refinement.

Finally, the audit deserves a place in the discussion rather than an apology. The evaluation stack that produced the leaked operating points had predeclared kill criteria, confidence intervals on every rate, frozen data splits, and guarded holdouts, and it still admitted materially lower reported false-alarm rates through test-informed threshold selection; because the affected evaluations differ in population as well as threshold provenance, the historical record is not a paired counterfactual and no single optimism factor is reported. The mechanism is not exotic and is not confined to this study. The community’s evaluation practice for learned warning methods would benefit from treating threshold provenance as a reportable property of every operating point.

7. Conclusions

On a synthetic benchmark validated against analytic limits and internal numerical checks, spanning three stability-failure archetypes, motion-only warning methods rank capsize risk well above chance at every forcing bandwidth tested, fail to transfer operating thresholds across failure modes, and discriminate event timing at chance inside an established severe regime; the majority of their false alarms coincide, in a mixed-generation diagnostic without a matched null, with high-envelope wave-group episodes. A test-selected evaluation had earlier reported materially lower false-alarm rates for the leading detector; the affected evaluations differ in population as well as threshold provenance, so the effect is reported qualitatively. The split-time experiments sharpen the same conclusion from inside the method: the upcrossing rate retained the more encouraging reliability pattern among the tested components, though no tested form passed a count-capture check, while the severity term that would time the event was, in 92 percent of heralded capsizes, not emitted with a ten-second opportunity before the recorded event. The natural-initial-condition experiment then argues the same point from the other side: within its balanced single-configuration design, with the arriving group’s engineered description fully known, capsize within the group is predicted at AUC 0.934 while the vessel’s entry state raises the point estimate by 0.004. That motivates, but does not price, a forward-sensing experiment: what was tested is a complete, noiseless, exactly aligned episode representation, and whether any realizable sensor recovers a useful fraction of it — under range limits, directional ambiguity, and estimation error (Nielsen et al. 2024; Lee and Kim 2025) — is the proposed next study. Further progress on the timing question appears to require information about the

incident wave field rather than refinement of motion-only architectures; shipboard sea-state and wave-field estimation is an active literature (Nielsen 2017; Nielsen et al. 2024; Lopac et al. 2026), with real-time wave-elevation reconstruction from measured motion now demonstrated in simulation and basin conditions (Zhang et al. 2026), and the benchmark’s frozen splits and audited protocol provide the evaluation discipline for that study, though an encounter-sensing experiment would require model extensions the benchmark does not yet contain.

Declarations

Data availability. The benchmark code, campaign definitions, preregistrations, and the complete numerical record, including corrected and superseded results retained as immutable history, will be public at github.com/wrobstory/rahola on publication.

Declaration of competing interest. The author declares no competing financial interests or personal relationships that could have influenced this work.

CRedit author statement. W. R. Story: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing, visualization.

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The benchmark code, campaign definitions, and the complete numerical record, including the pre-audit results retained as historical artifacts, are available from the author.

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